ISHU SINGH

GIS Engineer | Geospatial Software Developer | Remote Sensing | AI & Automation

Email: singhishu2060@gmail.com **LinkedIn:** linkedin.com/in/ishu-singh-geo **GitHub:** github.com/ishunteam-png
Portfolio: ishunteam-png.github.io **GIS Portfolio:** ishunteam-png.github.io/gis-portfolio (17 live dashboards) **Location:** Tbilisi, Georgia — Remote

5 Products Shipped | 2,655 InSAR Persistent Scatterers | 17 Live GIS Dashboards | 4 Hospital Portals | 7 Automation Workflows

PROFESSIONAL SUMMARY

Geospatial software developer and GIS engineer with production-proven experience in SAR/InSAR processing, satellite remote sensing, spatial data pipelines, and full-stack engineering. Self-taught the complete MintPy and SNAP InSAR workflow, then executed a 3-year Sentinel-1 analysis over Delhi road infrastructure — 2,655 Persistent Scatterers extracted, EGMS Level-3 equivalent vertical deformation output, subsidence rates spanning -10.3 to +4.7 mm/yr. Coupled the InSAR output with YOLOv8 road-damage segmentation to operationalise a satellite-to-pavement failure chain for infrastructure maintenance targeting.

Shipped three production systems currently in active use: an AI-assisted web platform, a Telegram-driven n8n automation suite with 7 live workflows, and a multi-tenant healthcare platform serving four hospitals with full audit logging. Built 17 independent geospatial research dashboards — each a live Leaflet application with full analysis pipeline — covering LULC classification, compound flood risk, mobility analytics, wildfire simulation, crop type fusion, air quality monitoring, vessel tracking, coastal erosion, and urban heat island analysis.

CORE COMPETENCIES

InSAR Processing | SAR Analysis | Persistent Scatterer Analysis | Satellite Remote Sensing | Land Use / Land Cover Classification | Geospatial Machine Learning | Spatial Data Engineering | Python Development | Full-Stack Web Development | GIS Development | Environmental Monitoring | Network Analysis | AI Integration | Workflow Automation | Sensor Fusion | Time Series Analysis | Cloud GIS | Data Pipeline Engineering

TECHNICAL SKILLS

Remote Sensing / SAR: Sentinel-1 / 2 / 5P | Landsat 8/9 | MintPy | SNAP | PyAPS3 | Google Earth Engine | LandTrendr | TROPOMI | VIIRS

GIS & Spatial Analysis: GeoPandas | rasterio | GDAL | OSMnx | NetworkX | Shapely | h3-py | pystac-client | xarray | OR-Tools | pvlib

Machine Learning / AI: scikit-learn | PyTorch | YOLOv8 (ultralytics) | OpenCV | DBSCAN | Random Forest | Sensor Fusion

Programming & Data: Python | TypeScript | JavaScript | React | DuckDB | pandas | pyarrow | scipy | NumPy

Visualization & Mapping: Leaflet.js | Chroma.js | Vite | Tailwind CSS | HTML / CSS

Automation & DevOps: n8n | Telegram Bot API | GitHub Actions | GitHub Pages

WORK EXPERIENCE

Freelance GIS Engineer & Remote Sensing Developer

Independent practice — client geospatial projects, SAR analysis, and spatial pipeline development

- Designed and executed a full InSAR processing chain from Sentinel-1 SLC acquisition through SBAS coregistration, PyAPS3 atmospheric correction, and MintPy time-series inversion — delivering EGMS Level-3 equivalent vertical deformation maps with 2,655 Persistent Scatterers over Delhi road corridors (subsidence -10.3 to +4.7 mm/yr).
- Integrated YOLOv8m semantic segmentation of road surface imagery with the InSAR PS dataset via spatial join: every road damage detection landed on a subsiding scatterer, establishing a satellite-to-pavement failure chain for infrastructure maintenance teams.
- Classified Bengaluru land cover from a 2020-2024 Sentinel-2 time series (OA 90.9%, kappa 0.89, built-up area +22.6%) and ran SAR-optical sensor fusion for Iowa crop type mapping — fusing Sentinel-1 and Sentinel-2 pushed accuracy from 86.3% to 92.7% (+6.4 pp) across 672 fields.

- Monitored 2,698 sq km of Amazonian deforestation via LandTrendr breakpoint detection on Google Earth Engine; tracked shoreline change at 15 Florida sites over a decade (Daytona Beach -27.4 m retreat, Miami Beach +24.0 m from federal renourishment).
- Built a multi-signal location suitability model across a 62,849-node pedestrian network (Spearman rho = 0.39 vs real cafe distribution); benchmarked OR-Tools GLS against greedy and Clarke-Wright — OR-Tools cut total drive time 20.3% across 60 stops and 5 vehicles.

Freelance Software Engineer & Automation Developer

Haimcore (AI platform) | Relief Guru (workflow automation) | Jalaram Healthcare

- Shipped Haimcore — a production AI-assisted web platform with release-versioned audit folders, rollback paths on every deployment, and idempotency guards preventing state corruption. Every LLM API boundary fails with the actual structured error, not a silent fallback.
- Built and deployed Relief Guru: a Telegram-driven n8n automation system with 7 workflows in production. Every external API call has typed error propagation; deployment is a single script with automated smoke tests and a one-step rollback. No manual steps in the release process.
- Developed and maintains Jalaram — a multi-tenant healthcare platform with 4 live hospital portals. Features include SystemLog audit trails for every clinical event, role-based access control, and real-time patient management dashboards. Processes live clinical data daily across all four sites.

GEOSPATIAL RESEARCH & PORTFOLIO PROJECTS

17 independent research projects — each with a live Leaflet dashboard and full analysis pipeline. All live at ishunteam-png.github.io/gis-portfolio

InSAR Road Subsidence — Delhi
Sentinel-1 | MintPy | SNAP | PyAPS3
3-year SAR stack | 2,655 PS | -10.3 to +4.7 mm/yr | EGMS-L3 output | live velocity dashboard

GeoAI Road Damage x InSAR
YOLOv8m | PyTorch | OpenCV
Road damage segmentation joined to InSAR PS layer | 100% of detections on subsiding scatterers

Sentinel-2 LULC — Bengaluru 2020-2024
pystac-client | scikit-learn | rasterio
OA 90.9% | kappa 0.89 | built-up +22.6% | 4-year epoch toggle

S1+S2 Crop Classification — Iowa
Sentinel-1 | Sentinel-2 | scikit-learn | USDA CDL
672 fields | S2-only 86.3% to fused 92.7% | +6.4 pp from SAR-optical fusion

NDVI Deforestation — Rondonia, Brazil
Google Earth Engine | Landsat 8/9 | LandTrendr
2,698 sq km cleared 2015-2024 | breakpoints show 2019 fire spike and 2024 policy reversal

Jakarta Compound Flood Risk
rasterio | scipy.ndimage | OSMnx
HAND + slope + drainage density + imperviousness | 1,360 schools + 138 hospitals mapped

15-Minute City — Paris
OSMnx | NetworkX | Shapely | alphashape
87.7% of hexes reach 5+ of 6 service categories | 47,812-node pedestrian network

Wildfire Spread — Park Fire CA 2024
Rothermel model | LANDFIRE | HRRR | VIIRS
961/1,024 cells burned by hour 120 | 28 m/h peak spread | VIIRS-validated perimeter

Urban Heat Island — Tokyo
Landsat 9 | Sentinel-2 | scipy
5.6 deg C LST delta Yamanote core vs suburban | Spearman rho = -0.82 with NDVI

Rooftop Solar Potential — Lisbon
py3dep | pvlib | PVGIS | OSMnx
12.5 GWh/yr across 450 rooftops | equivalent to 4,178 households

H3 Mobility Analytics — NYC Yellow Taxi

h3-py | DuckDB | pandas | pyarrow

57,000 daily trips in H3 hexes | AM commute-in / PM airport-dropoff asymmetry mapped

Sentinel-5P NO2 — Delhi

TROPOMI | xarray | CPCB

100% WHO exceedance Jan vs 10% Jun | winter inversion vs monsoon washout clearly separated

AIS Vessel Tracking — Singapore Strait

H3 res-8 | DBSCAN | MarineTraffic

978 vessels | 200 anchored in 8 DBSCAN clusters | +18% Red Sea diversion backlog

Coastal Erosion — Florida Atlantic

Sentinel-2 | NDWI | scikit-image | Shapely

Daytona Beach -27.4 m | Miami Beach +24.0 m | Hurricane Irma 2017 footprint visible

Cafe Location Intelligence — Tbilisi

OSMnx | GeoPandas | scipy | Leaflet

62,849-node network | 504 sq km | rho = 0.39 vs real cafe distribution

Advanced VRP — Tbilisi

OSMnx | NetworkX | OR-Tools

OR-Tools GLS vs Clarke-Wright | -20.3% drive time | 60 stops / 5 vehicles

Web GIS Dashboard — Delhi PS

Leaflet.js | Chroma.js | Vanilla JavaScript

~480-line single-file app | 6 presets | split view | URL-hash state persistence

EDUCATION

B.Sc. Geodesy and Geoinformatics Engineering

Georgian Technical University | Tbilisi, Georgia | Expected April 2026

- Core coursework: satellite positioning, remote sensing, GIS analysis, geodesy, cartography, and spatial data management.
- Developed 17 independent satellite analysis projects in parallel with the degree — InSAR, ML classification, routing, mobility, wildfire, solar, and climate domains. All live on GitHub Pages with interactive dashboards.